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POSTER

Are Deepfake Detectors Looking at the Right Face?

A Semantic Region Explanation Consistency Study

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When a deepfake detector decides, is it looking at the face — or at everything else? And do different architectures agree?

Motivation

Deepfake detectors reach 96–99% accuracy on FaceForensics++ yet lose 30–43 points on Celeb-DF. Saliency maps are routinely shown as evidence, but a heatmap only becomes forensic evidence once we know **which facial region** it covers, and whether other architectures point at the same place.

Method

1 · Detectors
8 ImageNet backbones (7 families) fine-tuned on 4,000 FF++ frames, 6 epochs

2 · Grad-CAM++
Map $S_d(x)$ for the predicted class of every test image

3 · Face parsing
BiSeNet → 6 regions: eyes, nose, mouth, skin, hair, background

4 · Attribution
Region share R, area-normalised E, and SAC between detectors

Relative attribution $R_{d,k}(x) = \Sigma\, S_d(x) \cdot M_k \,/\, \Sigma\, S_d(x)$ (rows sum to 100%)

Area-normalised $E_{d,k} = \bar{R}_{d,k} \,/\, \bar{a}_k$ ($E > 1$: more saliency per pixel than average)

SAC(d_1, d_2) = mean over images of $\cos(\, \mathbf{R}_{d_1}(x), \mathbf{R}_{d_2}(x)\,)$

Test: FF++ 2,000 frames (991 real / 1,009 fake), Celeb-DF v2 1,000 frames. All-zero Grad-CAM++ maps are excluded and reported. SAC is checked against pixel-level map correlation and Celeb-DF error overlap.

What the detectors look at



Grad-CAM++ (predicted class) on three FF++ test frames. Xception and ResNet50: broad, face-centred maps. EfficientNet-B4: the top-right corner on every image — a content-independent artifact. ViT-B/16: sparse maps, on the background (top) or small facial areas.

Key findings

1.3–3.2× their pixel share goes to eyes + nose + mouth in **every** detector — small raw shares are a size effect.

0.95–0.98 SAC within Xception · ResNet50 · Inc-ResNet-v2, with pixel-level r = 0.73–0.94. ViT-B/16 matches no CNN at pixel level ($r \le 0.16$).

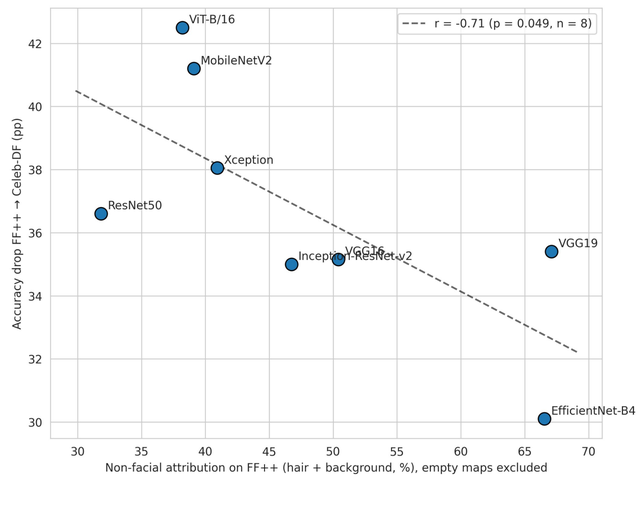
r = −0.71 non-facial share vs. FF++→Celeb-DF accuracy drop ($p = 0.049$, $n = 8$) — but not for AUC drop ($r = -0.34$). Exploratory.

Region attribution on FF++

Detector	Skin	Bg	Non-facial (E)	Features (E)	Empty %
Xception	48.3	25.6	40.9 (0.59)	10.8 (2.42)	0.00
EfficientNet-B4	27.7	54.1	66.5 (0.95)	5.7 (1.29)	0.00
ViT-B/16	47.9	25.9	38.2 (0.55)	13.9 (3.15)	54.65
ResNet50	54.9	20.2	31.8 (0.46)	13.2 (2.98)	0.00
VGG16	39.8	34.4	50.4 (0.72)	9.8 (2.21)	0.45
VGG19	27.3	48.3	67.1 (0.96)	5.6 (1.27)	0.60
MobileNetV2	49.2	27.2	39.1 (0.56)	11.7 (2.64)	45.20
Inception-ResNet-v2	43.4	33.6	46.7 (0.67)	9.8 (2.21)	0.00
Area share $\bar{a}_k$	25.8	48.6	69.7	4.44	–

Mean relative attribution R (%) over non-empty maps; area-normalised E in brackets. Features = eyes + nose + mouth; non-facial = hair + background. EfficientNet-B4 and VGG19 have $E \approx 1$ almost everywhere: diffuse maps, not a preference for background.

Explanations vs. generalisation



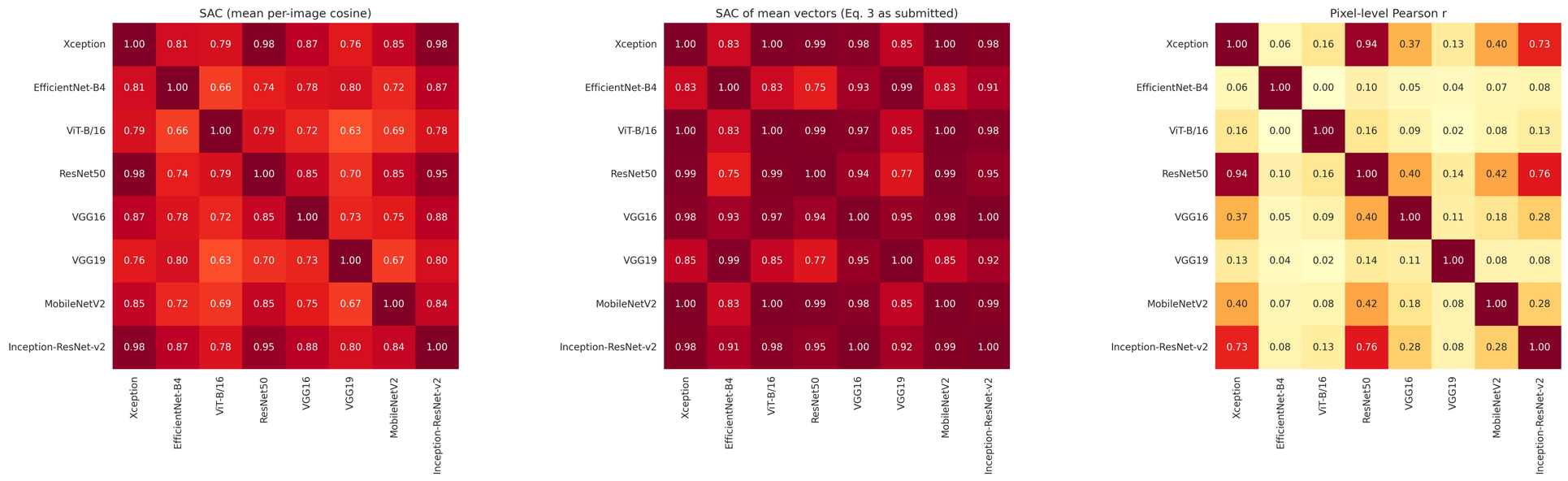
Non-facial attribution on FF++ (hair + background, %), empty maps excluded

Detectors with **more** non-facial attribution on FF++ lost **less** accuracy on Celeb-DF.

Pearson $r = -0.71$ ($p = 0.049$); Spearman $\rho = -0.69$; bootstrap 95% CI [−0.96, −0.27]; leave-one-out −0.82 to −0.52.

Absent for AUC drop ($r = -0.34$, $p = 0.42$) → possibly calibration shift. One transfer pair, $n = 8$: a hypothesis, not a robustness marker.

Do architectures agree? Semantic Attribution Consistency



Left: SAC. **Middle:** cosine of mean vectors (for contrast). **Right:** pixel-level Pearson r.

ViT's *average* allocation matches the CNNs (0.83–0.996), but per image it diverges.

Over 28 pairs, SAC tracks pixel agreement ($r = 0.83$) more than shared Celeb-DF errors ($r = 0.42$; Jaccard 0.31–0.54).